

Math 222: Problem Set 9

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Problem 1

In Exercises 1 to 8, sketch the given vector field or a small multiple of it.

2. $F(x, y) = (4, 0)$

Everywhere on the plane, there is a vector pointing towards the x-axis.

5. $F(x, y) = (2y, x)$

I will draw on my ipad.

In Exercises 15 to 18, show that the given curve $c(t)$ is a flow line of the given velocity vector field $F(x, y, z)$.

Problem 2

16. $c(t) = (t^2, 2t - 1, \sqrt{t}), t > 0; F(x, y, z) = (y + 1, 2, \frac{1}{2z})$

Solution

$$c'(t) = (2t, 2, \frac{1}{2\sqrt{t}}) = (2t, 2, \frac{1}{2 * (\sqrt{t})})$$

26. Sketch the gradient field $-\nabla V$ for $V(x, y) = (x + y)/(x^2 + y^2)$ and the equipotential surface $V = 1$.

I will do this on my tablet.

Problem 3